

ITERATED ENDPOINT-PEAK EXTRACTION ON PERMUTATIONS: ALL-RANK IMAGES, SECTIONS, AND A SHARP CLOCK

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ABSTRACT. Give both endpoints of a permutation a fictitious neighbour value zero, read all local-maximum values from left to right, and standardize the resulting word. Iterating this rule produces a deterministic variable-rank dynamics on the finite disjoint union of symmetric groups. We prove, for every source rank n and iterate $k \geq 1$, the exact image

$$P^k(\mathfrak{S}_n) = \bigsqcup_{1 \leq m \leq \lceil n/2^k \rceil} \mathfrak{S}_m.$$

The reverse inclusion is constructive: top values encode any target order, small values separate consecutive peaks, and unused values form a decreasing tail; composing minimal odd lifts gives a right section at every rank. It follows that the image has size $\sum_{m \leq \lceil n/2^k \rceil} m!$ and that the sharp maximum absorption time is $\lceil \log_2 n \rceil$, with a recursively lifted witness for every n . As a secondary result, we express every one-step target multiplicity as a sum of linear-extension counts of comparison-word posets augmented by the target peak-value order. An exhaustive exact audit covers all 409,113 permutations through rank nine; it tests the formulas but is not used as proof. Static zero-boundary peak distributions, ordinary pinnacle sets and admissible orders, and run-sorting equidistribution are treated as credited background.

1. A VARIABLE-RANK SELF-MAP

For a word of distinct values, std replaces its smallest value by 1, its next smallest by 2, and so on. Fix $N \geq 1$ and use the finite carrier

$$\mathfrak{S}_{\leq N} := \bigsqcup_{1 \leq n \leq N} \mathfrak{S}_n.$$

For $\pi = \pi_1 \cdots \pi_n$, set $\pi_0 = \pi_{n+1} = 0$. Let

$$\text{Peak}(\pi) := (\pi_i : \pi_{i-1} < \pi_i > \pi_{i+1})$$

in left-to-right order, and define

$$(1) \quad P(\pi) := \text{std}(\text{Peak}(\pi)).$$

The global maximum is always selected, so the output is nonempty. For example,

$$\pi = 4, 1, 6, 2, 5, 3, \quad \text{Peak}(\pi) = 4, 6, 5, \quad P(\pi) = 1, 3, 2.$$

Thus P is a literal self-map of $\mathfrak{S}_{\leq N}$, not of a single $\mathfrak{S}_n$. Because N is arbitrary, the rank-wise theorems are stated for every n ; inside a fixed carrier they are read with $1 \leq n \leq N$.

We call $\text{Peak}(\pi)$ the *endpoint-inclusive peak-value word*. It is not literally the conventional pinnacle set, whose peaks are restricted to interior positions. Fu uses the exact convention $\pi_0 = \pi_{n+1} = 0$ for static peak/double-descent distributions [7]; all such static distribution results receive zero contribution credit. The bridge to ordinary pinnacle theory is the padding

$$\pi_1 \cdots \pi_n \mapsto 1, (\pi_1 + 2), \dots, (\pi_n + 2), 2.$$

Its ordinary interior peak values, in left-to-right order, are precisely the entries of $\text{Peak}(\pi)$ shifted by two. Thus ordinary pinnacle sets and their admissible orders are nearest background rather than literal same-convention owners [2–6, 8]. Alexandersson–Nabawanda construct an auxiliary bijection giving a multivariate equidistribution between input peak-value sets and peak-value sets after run-sorting; run-sorting itself is not a pointwise invariant map [1]. Static peak/pinnacle distributions, admissible orders, fixed-set counts, that equidistribution, and generic zigzag-poset technology all receive zero contribution credit. The lead result below is instead the ordered zero-boundary peak-value word followed by standardization and iteration, its images at every iterate, and explicit right sections.

2. ONE-STEP PACKING AND A RIGHT SECTION

No two peak positions are adjacent. Hence every $\pi \in \mathfrak{S}_n$ satisfies

$$(2) \quad 1 \leq |P(\pi)| \leq \left\lceil \frac{n}{2} \right\rceil.$$

Since $\lceil n/2 \rceil < n$ for $n > 1$, every nonsingleton step strictly decreases rank; equality in the displayed packing bound may occur.

Fix $\sigma = \sigma_1 \cdots \sigma_m \in \mathfrak{S}_m$ and $n \geq 2m - 1$. Define high values $h_i = n - m + \sigma_i$ and the lift

$$(3) \quad L_{n,m}(\sigma) := h_1, 1, h_2, 2, \dots, h_{m-1}, m-1, h_m, n-m, n-m-1, \dots, m,$$

where the terminal decreasing block is omitted if $n-m < m$. The high values are exactly $n-m+1, \dots, n$; the displayed valleys use $1, \dots, m-1$; and the tail contains every remaining value. Thus (3) is a permutation of $[n]$.

Lemma 1 (one-step section). *For every $\sigma \in \mathfrak{S}_m$ and $n \geq 2m - 1$,*

$$(4) \quad P(L_{n,m}(\sigma)) = \sigma.$$

Proof. The first high value exceeds the fictitious left boundary and its following valley. Every interior high lies between smaller valleys. The last high exceeds its left valley and either the first tail value or the fictitious right boundary. Hence all highs are peaks. A valley has a larger high neighbour, and every tail entry has a larger left neighbour, so no low value is a peak. The selected word is $h_1 \cdots h_m$, whose standardization is σ . $\square$

Already, (2) and Lemma 1 show that the one-step image consists of every permutation of each rank at most $\lceil n/2 \rceil$. The section is value-explicit rather than a counting argument.

3. EVERY ITERATE IMAGE AND THE SHARP CLOCK

Theorem 2 (all-rank images and sections). *For every $n, k \geq 1$,*

$$(5) \quad P^k(\mathfrak{S}_n) = \bigsqcup_{1 \leq m \leq \lceil n/2^k \rceil} \mathfrak{S}_m, \quad |P^k(\mathfrak{S}_n)| = \sum_{m=1}^{\lceil n/2^k \rceil} m!.$$

Moreover, every target in the displayed union has an explicit right section obtained by composing the lifts (3).

Proof. Iterating (2) gives

$$|P^k(\pi)| \leq \left\lceil \frac{1}{2} \left\lceil \cdots \left\lceil \frac{n}{2} \right\rceil \cdots \right\rceil \right\rceil = \left\lceil \frac{n}{2^k} \right\rceil,$$

which proves the forward inclusion.

For the reverse inclusion, fix $\sigma \in \mathfrak{S}_m$ under this rank bound. The integer inequality $m \leq \lceil n/2^k \rceil$ is equivalent to

$$(6) \quad n \geq 2^k m - (2^k - 1).$$

Put $a_k = m$ and recursively $a_j = 2a_{j+1} - 1$ for $j = k-1, \dots, 1$. Starting with $\sigma_k = \sigma$, set

$$\sigma_j = L_{a_j, a_{j+1}}(\sigma_{j+1}) \quad (j = k-1, \dots, 1), \quad \pi = L_{n, a_1}(\sigma_1).$$

Condition (6) is exactly what makes the outer lift legal; all inner lifts use their minimal odd length. Lemma 1 gives $P(\pi) = \sigma_1$ and $P(\sigma_j) = \sigma_{j+1}$, hence $P^k(\pi) = \sigma$. This proves the reverse inclusion constructively. The cardinality follows because different ranks are disjoint in the carrier. If some $a_j = 1$, the corresponding $L_{1,1}$ is the identity, so the construction also covers arbitrarily large iterate ranks. $\square$

Let $\tau(\pi)$ be the first time the orbit reaches the one-letter permutation.

Theorem 3 (sharp clock). *The singleton is the unique recurrent state. Every $\pi \in \mathfrak{S}_n$ satisfies $\tau(\pi) \leq \lceil \log_2 n \rceil$, and for every $n \geq 1$,*

$$(7) \quad \max_{\pi \in \mathfrak{S}_n} \tau(\pi) = \lceil \log_2 n \rceil.$$

There is a recursively defined equality witness for every n .

Proof. For $n > 1$, (2) strictly decreases rank, proving the recurrent claim and the upper bound in (7). Define $w_1 = 1$ and, for $n > 1$,

$$(8) \quad w_n := L_{n, \lceil n/2 \rceil}(w_{\lceil n/2 \rceil}).$$

The lift is legal because $n \geq 2\lceil n/2 \rceil - 1$, and Lemma 1 gives $P(w_n) = w_{\lceil n/2 \rceil}$. Induction between consecutive powers of two now yields $\tau(w_n) = 1 + \tau(w_{\lceil n/2 \rceil}) = \lceil \log_2 n \rceil$. $\square$

Theorem 2, rather than the fibre formula below, is the primary axis: it proves both inclusions at every iterate and supplies a target-level inverse, not only an image census.

4. A TARGET-RESOLVED ONE-STEP MULTIPLICITY

For $n > 1$, a comparison word $w = w_1 \cdots w_{n-1} \in \{U, D\}^{n-1}$ records U when $\pi_i < \pi_{i+1}$ and D otherwise. Its endpoint-inclusive peak positions are

$$(9) \quad \{1 : w_1 = D\} \cup \{i : 1 < i < n, w_{i-1} = U, w_i = D\} \cup \{n : w_{n-1} = U\}.$$

For $n = 1$, the empty comparison word has the unique position as a peak.

Let $p_1 < \cdots < p_m$ be the positions in (9). The adjacent-comparison poset has relations

$$(10) \quad i <_w i+1 \quad (w_i = U), \quad i+1 <_w i \quad (w_i = D).$$

For $\sigma \in \mathfrak{S}_m$, adjoin the peak chain

$$(11) \quad p_{\sigma^{-1}(1)} < p_{\sigma^{-1}(2)} < \cdots < p_{\sigma^{-1}(m)}$$

and denote the resulting poset by $Q(w, \sigma)$. Every base peak is maximal under (10), so adjoining (11) creates no cycle. Write $e(Q)$ for the number of linear extensions of Q .

Theorem 4 (complete one-step fibre). *For every $\sigma \in \mathfrak{S}_m$,*

$$(12) \quad |P^{-1}(\sigma) \cap \mathfrak{S}_n| = \sum_{\substack{w \in \{U, D\}^{n-1} \\ w \text{ has } m \text{ endpoint-inclusive peaks}}} e(Q(w, \sigma)).$$

If $m > \lceil n/2 \rceil$, the sum and the fibre are both empty.

Proof. Fix w . A linear extension of $Q(w, \sigma)$, read from smaller to larger, assigns the ranks $1, \dots, n$ to positions. Relations (10) force exactly the comparison word w , while (11) forces the left-to-right peak-value word to standardize to σ . Conversely, a permutation with comparison word w and extracted target σ orders its positions by increasing value and gives one such linear extension. This is a bijection for each w . Every source permutation has a unique comparison word, so summing the disjoint classes proves (12). The rank boundary follows from (2); the case $n = m = 1$ has one empty word and one extension. $\square$

Formula (12) uses standard poset technology and is deliberately a secondary inverse atlas. It does not reclaim static pinnacle-set or admissible-order results.

5. EXACT AUDIT AND CONCLUSION

An independent standard-library verifier enumerates every permutation through rank nine. It compares full iterate image *sets* through five steps, constructs a right section for every feasible target through rank eight, checks that (8) saturates every intermediate packing bound, and compares every target fibre through rank eight with an independent subset-DP linear-extension count. All 1,228,181 exact assertions pass. Table 1 is frozen counterexample pressure, not proof evidence.

TABLE 1. Exact source-rank profile.

n	1	2	3	4	5	6	7	8	9
$ \mathfrak{S}_n $	1	2	6	24	120	720	5040	40320	362880
$ P(\mathfrak{S}_n) $	1	1	3	3	9	9	33	33	153
$\max \tau$	0	1	2	2	3	3	3	3	4

Repeated endpoint-peak extraction therefore has a complete rank geometry: packing gives every upper boundary, and explicit lifts fill every rank below it. The same lifts provide equality witnesses for the sharp clock, while the comparison-poset sum resolves one-step multiplicities target by target.

LIMITATIONS

The theorem uses positive permutations, boundary value zero, and standardization after each extraction. Different endpoint conventions, words with ties, cyclic peaks, or unstandardized value dynamics are outside scope. Fu’s exact-convention static distributions, ordinary pinnacle theory under the padding bridge, run-sorting equidistribution, and generic zigzag-poset methods receive zero contribution credit. The primary-source audit was bounded, and no novelty or priority claim is made. Finite enumeration cannot replace the proofs. The manuscript remains under `HOLD_EXTERNAL`.

DATA AVAILABILITY

No external data were used. The anonymous artifact contains the standard-library verifier and frozen exact output. External release of the artifact is not authorized at this stage.

ETHICS STATEMENT

The work uses no human participants, animals, personal data, or deployed decision system.

AUTHOR CONTRIBUTIONS

Contributor identities are withheld for anonymous review. The anonymous author team takes responsibility for conceptualization, formal analysis, software, validation, and writing.

CONFLICT OF INTEREST

The authors declare no known conflict of interest relevant to this internal mathematical study.

FUNDING

No funding claim is made in this anonymous internal draft.

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